

Fei Wu

E-mail: fw407@exeter.ac.uk Homepage: <https://fei407.github.io/>

EDUCATION

-
- | | |
|---|------------------------|
| University of Exeter, United Kingdom | Oct. 2023 – Present |
| Ph.D. in Computer Science | |
| • Fully funded by the China Scholarship Council and University of Exeter (CSC-Exeter) | |
| University of Electronic Science and Technology of China (UESTC), China | Sept. 2019 – July 2022 |
| M.Eng. in Electronic and Communication Engineering | |
| • Ranked 23rd out of 332 students (Top 6.9%); awarded the Outstanding Graduate of Sichuan Province | |
| Chengdu University of Technology (CDUT), China | Sept. 2015 – July 2019 |
| B.Eng. in Electronic and Information Engineering | |
| • Ranked 1st out of 135 students (Top 0.7%); admitted to UESTC (M.Eng.) through direct recommendation | |

EXPERIENCE

-
- | | |
|--|----------------|
| 1. University of Exeter – Postgraduate Research Assistant (Computer Science) | 2025 – present |
| • Conducted research on efficient fine-tuning and federated learning systems | |
| 2. Chengdu RuiXinXing Co., Ltd. – FPGA Engineer (SLAM) | 2022 – 2023 |
| • Investigated efficient algorithm-hardware co-design solutions for mobile robotics SLAM | |

RESEARCH PUBLICATIONS

-
- [1] **F. Wu**, J. Hu, G. Min, S. Wang, “[Efficient Orthogonal Fine-Tuning with Principal Subspace Adaptation](#),” *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, April, 2026.
- [2] **F. Wu**, J. Hu, G. Min, S. Wang, “[Adaptive Rank Allocation for Federated Parameter-Efficient Fine-Tuning of Language Models](#),” *IEEE Transactions on Computers (IEEE TC)*, Accepted, Jan. 2026.
- [3] Y. Liu, **F. Wu**, N. Zhao et al., “[NVP: A Flexible and Efficient Processor Architecture for Accelerating Diverse Computer Vision Tasks including DNN](#),” *IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II)*, vol. 70, no. 1, pp. 271-275, Jan. 2023.
- [4] **F. Wu**, N. Zhao, Y. Liu et al., “[A Review of Convolutional Neural Networks Hardware Accelerators for AIoT Edge Computing](#),” *The International Conference on UK-China Emerging Technologies (UCET)*, 2021, pp. 71-76.
- [5] Y. Liu, J. Xiao, **F. Wu** et al., “[A Fast and Efficient FPGA-based Level Set Hardware Accelerator for Image Segmentation](#),” *Journal of Electronics and Information Technology (JEIT)*, vol. 32, no. 6, pp. 1525-1532, 2021.

RESEARCH PROJECTS

-
- EU Horizon Europe, REFINE: Real-time Fine-grained Air Quality Monitoring with Intelligent and Robust Multi-UAV Networks. (PhD Researcher, Grant No. 101129910)
 - EPSRC, New Horizons, Real-Time Federated Learning at the Wireless Edge via Algorithm–Hardware Co-Design. (PhD Researcher, EPSRC Grant EP/X019160/1)
 - EU Horizon Europe, ASCENT: Autonomous Vehicular Edge Computing and Networking for Intelligent Transportation. (PhD Researcher, Grant No. 101086159)
 - NSFC-CAEP, Joint Fund Project, AI Brain Models and Hardware Architectures for Multi-Task Learning in Complex Environments. (Main Research Participant, Grant No. U2030204)

INDUSTRIAL PROJECTS

-
- Pedestrian Tracking & Following for Robots, Chengdu RuixinXing Technology Co., Ltd. 2022 – 2023
 - Developed a LiDAR-vision fusion method enabling single-target tracking with **> 95%** success rate
 - Implemented an end-to-end system on NVIDIA Jetson Xavier **at 30 FPS** with 15 W power consumption
 - NVP: Neural Visual Processor**, SenseTime Co., Ltd. 2020 – 2022
 - Designed a multi-core architecture with segmented ring-bus interconnect, achieving **>80% PE utilization**
 - Verified single-core prototype on FPGA, demonstrating **3× energy efficiency improvement**
 - Conducted PPA evaluation in 40 nm CMOS technology, with **73.81mm²** area and power consumption **< 2 W**
 - FPGA Accelerator for MRI Segmentation, West China Hospital of Sichuan University 2019 – 2020
 - Designed a parallel FPGA architecture for level-set segmentation, **achieving 10.7× speedup with 2.2 W**
 - Implemented UART interface and FPGA-based demo
 - Weightel: Vehicle Intelligent Weighing System, Griffith-Elder Co., Ltd. & Camrong Co., Ltd. 2020 – 2022
 - Designed in-vehicle weighing and aggregation nodes for calibration and multi-node data fusion
 - Developed CAN-bus communication protocol, supporting up to 12 measurement nodes
 - Integrated and validated system in real-world environments, achieving **dynamic weighing error < 0.1%**

PATENTS

- [1] J. Zhou, L. Chang, L. Zhou, **F. Wu**, Data Processing Method and Apparatus, Computer Device, and Storage Medium PCT/CN2021/115789 (WO2022179075), China Patent CN202110221235.3 (Granted)
- [2] J. Zhou, L. Chang, L. Zhou, **F. Wu**, Image Processing Method and Apparatus, Computer Device and Storage Medium PCT/CN2021/115745 (WO2022160704), China Patent CN202110132579.7 (Granted)
- [3] J. Zhou, L. Chang, L. Zhou, **F. Wu**, Data Processing Method and Apparatus, Computer Device, and Storage Medium PCT/CN2021/115799 (WO2022160706); China Patent CN202110132573.X (Granted)

AWARDS & HONORS

Competitions

- National Third Prize, The China Graduate Circuit Design Contest 2021
- National Third Prize**, The China Graduate Electronics Design Contest (**Top 8.3%**) 2021
- National Second Prize**, The China Graduate Electronics Design Contest (**Top 3.7%**) 2020
- National Second Prize, Huawei Special Competition, The China Graduate Electronics Design Contest 2020

Honors

- Outstanding Graduate of Sichuan Province, China (**nominated from Top 32** in the Department) 2022
- Outstanding Graduate of University of Electronic Science and Technology of China 2022

Scholarships

- CSC-Exeter Scholarship and the additional Limetree Capital PhD Scholarship** 2023 – 2027
- First-Class Academic Scholarship of University of Electronic Science and Technology of China 2021
- Outstanding Student Scholarship of Chengdu University of Technology 2016 – 2018

SERVICES

Reviewer

- Conference: **ICML'26, ICLR'26, AAAI'26**, ICDCS'24, TrustCom'25, HPCC'25, ICESS'25, IUCC'24
- Journal: IEEE TPDS, IEEE TC, IEEE IoTJ, IEEE TAI

Supervision

Mentored an undergraduate student ([Harry Min](#)) from the University of Cambridge on a summer research project at Department of Computer Science, University of Exeter (Aug. – Sept. 2024), under the supervision of [Prof. Jia Hu](#).

SKILLS

Certificates	AMD Training – Accelerating Your Application with AMD GPUs NVIDIA Training – Efficient Large Language Model (LLM) Customization
Technical Skills	<u>Programming Languages</u> : Python , C, C++, Verilog , SystemVerilog, VHDL <u>Hardware & Embedded Platforms</u> : NVIDIA Jetson AGX Orin, Orin Nano Super , Raspberry Pi 5, AMD Zynq-7035, VC707, ZCU104, Atmel ATSAME51, STM32 <u>Frameworks & Libraries</u> : Docker, Hugging Face Transformers, PEFT , Flower, Pytorch <u>Tools & Platforms</u> : Linux (x86 & ARM), Slurm , Vivado, MATLAB, PyCharm, VS Code
Language Skills	Native Mandarin speaker and fluent in English